

Systematic Characterisation of Velocity-Curvature Power Law Analysis Protocols Across Biologically Informed Parameter Space

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Abstract

Lacquaniti et al.'s (1983) velocity–curvature power law (exponent $\beta \approx 1/3$) is widely regarded as a fundamental property of biological movement, with divergences from this value proposed as sensitive motor biomarkers. Exploiting these biomarkers requires systematic characterisation of how β is calculated. We conducted a pre-registered simulation spanning 18 million biologically informed parameter combinations across six analytical pipelines, validated against seven empirical datasets. The simulation revealed systematic compression of β in the presence of noise, reflecting a mismatch between embodied, fractional kinematics and the integer-order numerical differentiation used in standard analyses. We recover underlying exponents in the range $\beta \approx 0.40$ – 0.49 . The widely reported $\beta \approx 1/3$ emerges only under restricted noise conditions where pipeline compression masks the underlying exponent. These findings indicate that the commonly reported exponent is largely a calculation artefact, highlighting limitations of numerical differentiation in analysing noisy biological signals.